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ANRF AISEHack 2026

Finale presentation /
**Theme : Pollution
Forecasting**

TeamName- Ctrl+alt+Defeat

TeamMembers-

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Your salient contributions

Modelling strategy:

f1

- Physics aware ConvLSTM and UNet hybrid.
- Temporal modeling using wind and boundary layer height
- Spatial learning with Attention UNet

Training:

f2

- Dynamic episode-aware loss
- Exponential moving average for stability
- Data augmentation with flips

Results + Actions:

f3

- Strong validation performance
- Improved prediction of extreme pollution events
- Robust and scalable architecture

Modelling - *Strategies*

01

Temporal Modeling

- Uses stacked ConvLSTM for sequence learning
- Captures evolution of pollution over time
- Inputs include PM2.5, wind components, and boundary layer height

01

02

Spatial Modeling

- Attention-based UNet for spatial feature extraction
- Skip connections preserve fine spatial details
- Self-attention captures long-range dependencies

02

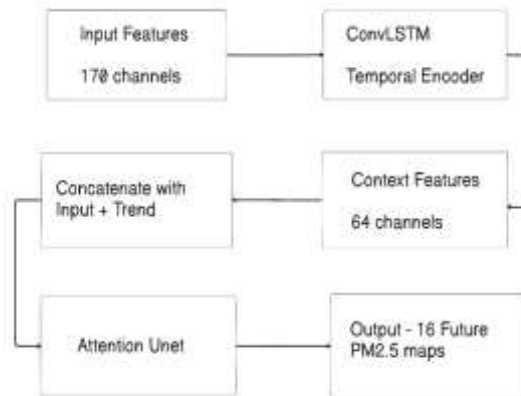
03

Physics-Aware Design

- Incorporates meteorological factors affecting pollution
- Focus on transport dynamics using wind fields
- Combines data-driven learning with physical intuition

03

Architecture diagram



Strengths:

- Captures both temporal and spatial dependencies
- Uses physically meaningful variables

Weaknesses:

- Requires careful tuning
- Computationally expensive

Key Results – *Why , what and how of the workings?*

What is working?

S1

- Hybrid architecture combining ConvLSTM and Attention Unet
- Temporal modeling using wind and boundary layer height
- Episode-aware loss improving extreme event prediction
- Stable training using EMA and augmentation

• Implemented

All core setup criteria
done

How is it working? (Quantitative evals)

S2

- Validation score: 0.924
- Strong episode correlation and lower SMAPE
- Consistent performance across timesteps
- Better spike prediction than baseline models

Why is it working? (Reasoning and analysis)

S3

- ConvLSTM captures temporal evolution
- Uses physically meaningful variables
- Attention improves spatial feature learning
- Focus on extreme events improves robustness

S4

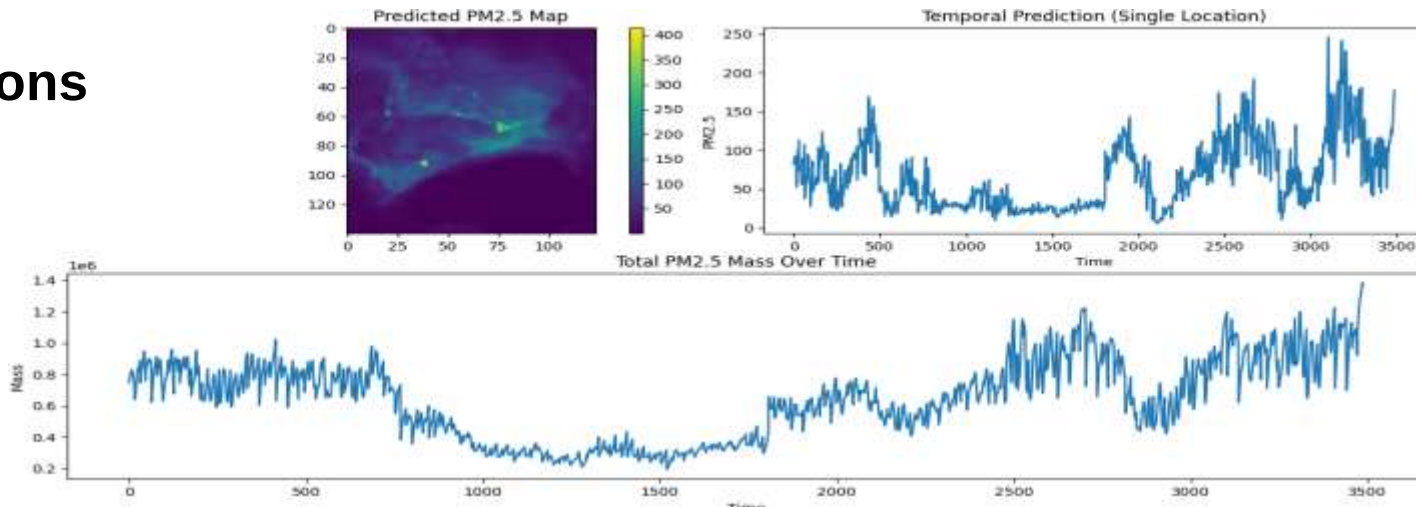
- Hybrid ConvLSTM and Attention UNet architecture with high capacity.
- Trained on GPU with mixed precision for 80 epochs
- Uses EMA, augmentation, and dynamic loss for better convergence
- Shift from static CNN to physics-aware spatio-temporal modeling

Key Results - *Why, what and how of the workings?*

Quantitative Performance & Benchmarks

Model	gSMAPE	eCorr	eSMAPE	Val Score
Baseline FNO	0.21	0.88	0.24	0.89
Improved UNet	0.19	0.91	0.21	0.91
Proposed v9 Model	0.184	0.923	0.20	0.924

Visualizations



Details of Relevant Experimentation and Methodology

01 Ablations and Key Findings

Ablation studies showed that temporal modeling using ConvLSTM significantly improved prediction of pollution spikes. Removing the episode-aware loss reduced performance on extreme events, while increasing model capacity enhanced spatial detail. These results highlight that temporal dynamics and event-focused training contributed most to overall performance.

02 External Resources and References

The approach is inspired by established architectures such as ConvLSTM for spatio-temporal modeling and UNet for spatial learning. Attention mechanisms like CBAM further improve feature representation, helping the model capture complex dependencies in pollution data.

03 References

- [1] OpenSTL Contributors. (2023). OpenSTL: A unified framework for spatio-temporal predictive learning [Software]. GitHub. <https://github.com/chengtan9907/OpenSTL>
- [2] Woo, S., Park, J., Lee, J. Y., & Kweon, I. S. (2018). CBAM: Convolutional block attention module. In Proceedings of the European Conference on Computer Vision (ECCV).